

Review retrospective:  
*multivariate Gaussian integrability, and a redundant import the  
job-count exposed*

Claude Opus 4.8 (1M ctx), reviewer GPT-5.5 Pro

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### Abstract

A soundness review of `Laplace/Multi/GaussianDomination.lean`: the four multivariate Gaussian integrability lemmas — linearly-tilted density, pure density, and the first- and second-moment integrands  $u_i e^{-c \sum u^2}$  and  $u_i u_j e^{-c \sum u^2}$  — that feed the multivariate integration-by-parts. Result: **a clean fidelity pass with one landed edit**, a redundant `import Laplace.Multi.Basic` removed under rebuild-arbitration, with the build’s own job count confirming the redundancy.

## 1 Setting

This is the integrability layer for the primer’s Stage 3: before one can compute  $\int u_i u_j e^{-Q(u)} du = Z(H^{-1})_{ij}$  by parts, one needs that integrand (and its lower-moment relatives) to be integrable. The file proves exactly those four facts on  $\iota \rightarrow \mathbb{R}$ , and it is the last small `Multi/` file before the large `Covariance*` trio. Its only consumer is `GaussianIBP` (itself reviewed earlier).

## 2 What was reviewed

Four public theorems (linear-tilt, pure Gaussian, first moment, second moment) and three private helpers (`sum_pi_single_mul`, the domination  $|x| \leq e^x + e^{-x}$ , and continuity of the Gaussian). The strategy, documented as following a GPT-5.5-Pro Phase-2 consult: prove the linear-tilt lemma from Mathlib’s *complex* `Gaussian GaussianFourier.integrable_cexp_neg_mul_sum_add` via `Integrable.norm`, then dominate the moment integrands by sums of linearly-tilted Gaussians (built with `Pi.single` tilts) using  $|x| \leq e^x + e^{-x}$ .

## 3 Fidelity / soundness findings

**Sound; the theorem statements are faithful.** The complex-to-real norm bridge (the cast identity plus `Complex.ofReal_exp/Complex.norm_real`) is correct; the `Pi.single` tilt constructions realise all of  $\pm u_i$  and  $\pm u_i \pm u_j$  (including  $i = j$ ); the dominations  $|x| \leq e^x + e^{-x}$  and  $|u_i u_j| \leq (e^{u_i} + e^{-u_i})(e^{u_j} + e^{-u_j})$  both hold as proved; and  $c > 0$  is correctly a hypothesis.

One observation worth recording, though it is *not* a file defect: for *empty*  $\iota$ , the unique point gives  $\sum u_k^2 = 0$ , so the integrands collapse to the constant 1 on a one-point space and are integrable for every  $c$ , not only  $c > 0$ . So  $c > 0$  is sufficient-but-not-strictly-necessary in that degenerate case. The file claims only sufficiency (“for  $c > 0$ , ... is integrable”) and makes no necessity assertion, so

nothing needs changing — but it is a clean reminder that “the hypothesis is necessary” and “the hypothesis is sufficient” are different claims, and only the latter is what an integrability lemma states.

## 4 Simplifications applied

**One landed: a redundant import.** The file imported `Laplace.Multi.Basic` but uses none of its definitions (no `partitionFunction/gibbsExpectation/gibbsCov`); the product-Lebesgue measure on  $\iota \rightarrow \mathbb{R}$  and the `Integrable` infrastructure all arrive through `Mathlib....Gaussian.FourierTransform`. Removed under rebuild-arbitration — and here the rebuild gave an unusually crisp confirmation: the lake build job count dropped from 2734 to 2726, i.e. eight modules left the build closure, modules that were present *only* via the removed import and are not needed. A load-bearing import would instead have failed elaboration. All seven declarations stayed byte-identical; GPT-affirmed; fast-forward-merged. This is a leaf file (only `GaussianIBP` imports it, and that imports `Basic` directly), so the single-file rebuild was a sufficient test and downstream is untouched.

GPT’s other eight proposals were declined: shorten or eliminate `sum_pi_single_mul` and `h_sum_pair`, replace the `apply h.congr; filter_upwards` blocks with `simp`, inline `continuous_gaussian`, use a standard `||Complex.exp||` lemma, shrink the `hG` simp sets, factor the four sign-cases through a helper, and rename privates. All are proof-golf, helper-extraction, or renaming — no dead code, no statement benefit — and were left alone.

## 5 Disagreements and open questions

None on the mathematics.

## 6 Lessons

- **The build’s job count is a free redundancy signal.** When removing an import drops modules from the closure (here  $2734 \rightarrow 2726$ ) and the build still passes, that is positive evidence the import was pulling in genuinely-unused transitive dependencies — a sharper confirmation than “it still compiles” alone. A load-bearing import cannot leave the closure without breaking the build.
- **Sufficient  $\neq$  necessary, and an integrability lemma only ever claims the former.** The empty-index degeneracy is a tidy reminder to read a hypothesis as “this suffices”, not “this is required”, when judging fidelity — the file was right, and a reviewer asking “is  $c > 0$  necessary?” is asking a strictly stronger question than the file answers.

## 7 Follow-ups

- The small/medium `Multi/` files are now reviewed (`Basic`, `AffineBilinearCov`, `QuadraticApprox`, `GaussianDomination`; `GaussianIBP` earlier). Remaining: `RescaledIntegrals.lean` (1547 lines, 47 lemmas — large but perhaps reviewable in one pass) and the `Covariance*` trio (3998–19611 lines, needing a sectioned approach).